PCET-NMVPM's

Nutan College of Engineering and Research,Talegaon, Pune

Department of Computer Science and Engineering(Artificial Intelligence)

Department: CSE -AI | Academic Year: 2025-2026

Semester: II | Year: **T**E | Date: **3**/**03**/2026

Mid Sem Examination Practice Questions

Advanced Machine Learning

Q.N	Question				Marks
1	Which of the following is a key characteristic of unsupervised learning?				1
	(a) Labeled data	(b) No labeled data	(c) Reinforcement signals	(d) Supervised feedback	
2	The primary goal of clustering is to:				1
	(a) Predict future values	(b) Group similar data points	(c) Minimize classification error	(d) Maximize regression accuracy	
3	K-means clustering minimizes:				1
	(a) Manhattan distance	(b) Euclidean distance squared	(c) Cosine similarity	(d) Hamming distance	
4	Which function in Scikit-learn is commonly used to apply K-means?				1
	(a) fit_transform ()	(b) fit_predict ()	(c) fit () and predict ()	(d) transform ()	
5	K-medoids differs from K-means because:				1
	(a) It uses centroids	(b) It uses actual data points as cluster centers	(c) It requires labeled data	(d) It minimizes regression error	
6	The first principal component captures:				1
	(a) Minimum variance	(b) Maximum variance	(c) Random variance	(d) Equal variance	
7	In hierarchical clustering, the bottom-up approach is also known as:				1
	(a) Divisive	(b) Agglomerative	(c) Partitioning	(d) Density-based	
8	Which of the following is NOT a clustering algorithm?				1
	(a) K-means	(b) K-medoids	(c) PCA	(d) Hierarchical clustering	
9	PCA is primarily used for:				1
	(a) Classification	(b) Regression	(c) Dimensionality reduction	(d) Clustering	
10	Choosing the optimal number of features in PCA often involves:				1
	(a) Silhouette score	(b) Scree plot / explained variance	(c) Confusion matrix	(d) ROC curve	
11	Eigenfaces are derived using:				1
	(a) K-means clustering	(b) PCA on facial images	(c) Hierarchical clustering	(d) K-medoids	
12	Which clustering method follows a top-down approach?				1
	(a) Agglomerative clustering	(b) Divisive clustering	(c) K-means clustering	(d) K-medoids clustering	
13	Which mathematical concept is central to PCA?				1
	(a) Eigenvalues and eigenvectors	(b) Gradient descent	(c) Decision trees	(d) Neural networks	
14	In the K-means algorithm, the "K" refers to:				1
	(a) Number of features	(b) Number of clusters	(c) Number of iterations	(d) Number of centroids per cluster	
15	K-means clustering minimizes:				1

	(a) Measure regression accuracy	(b) Determine optimal K in clustering	(c) Evaluate classification precision	(d) Reduce dimensionality	
16	The first principal component captures:				1
	(a) Minimum variance	(b) Maximum variance	(c) Random variance	(d) Equal variance	
17	The Silhouette score is used to:				1
	(a) fit_transform ()	(b) fit_predict ()	(c) fit () and predict ()	(d) transform ()	
18	In classification of LFW (Labeled Faces in the Wild) images, PCA helps by:				1
	(a) Extracting eigenfaces for feature representation	(b) Directly labeling images	(c) Performing clustering only	(d) Increasing image resolution	
18	Which type of learning does not use labeled data?				1
	a) Supervised Learning	b) Reinforcement Learning	c) Unsupervised Learning	d) Semi-supervised Learning	
19	Which of the following is a common task in unsupervised learning?				1
	a) Classification	b) Regression	c) Clustering	d) Prediction	
20	Clustering is mainly used to				1
	a) Predict outputs	b) Group similar data points	c) Label datasets	d) Train neural networks	
21	K in K-Means represents				1
	a) Number of features	b) Number of clusters	c) Number of iterations	d) Number of datasets	
22	K-Means clustering is an example of				1
	a) Hierarchical clustering	b) Flat clustering	c) Density clustering	d) Supervised learning	
23	Which distance measure is commonly used in K-Means?				1
	a) Manhattan Distance	b) Euclidean Distance	c) Hamming Distance	d) Jaccard Distance	
24	The centroid in K-Means represents				1
	a) The farthest point in cluster	b) The mean of data points in cluster	c) The smallest value	d) Random point	
25	Which library provides KMeans implementation in Python?				1
	a) NumPy	b) TensorFlow	c) Scikit-learn	d) Pandas	
30	Which method helps determine the optimal number of clusters?				1
	a) Elbow Method	b) Silhouette Score	c) Gradient Descent	d) Linear Regression	
31	Silhouette score measures				1
	a) Cluster accuracy	b) Cluster separation and cohesion	c) Number of clusters	d) Data variance	
32	K-Medoids differs from K-Means because it uses				1
	a) Mean values	b) Median values	c) Actual data points as cluster centers	d) Random vectors	
33	K-Medoids is more robust to				1
	a) Missing values	b) Noise and outliers	c) Large datasets	d) High dimensions	
34	Hierarchical clustering builds clusters in the form of				1
	a) Graph	b) Tree	c) Matrix	d) Table	
35	The tree diagram used in hierarchical clustering is called				1
	a) Histogram	b) Scatter plot	c) Dendrogram	d) Heatmap	
36	Divisive clustering follows which approach?				1

	a) Bottom-up	b) Top-down	c) Random clustering	d) Density-based	
37	Agglomerative clustering follows which approach?				1
	a) Bottom-up	b) Top-down	c) Random clustering	d) Parallel clustering	
38	PCA is mainly used for				1
	a) Data classification	b) Data visualization	c) Dimensionality reduction	d) Data cleaning	
39	PCA converts correlated variables into				1
	a) Dependent variables	b) Independent variables	c) Principal components	d) Regression values	
40	Eigenfaces technique is used in				1
	a) Speech recognition	b) Face recognition	c) Image compression	d) Video streaming	
1	A supermarket wants to analyze customer purchases. They find that 40% of transactions include milk, 30% include bread, and 20% include both. Scenario Question: Calculate the support and confidence for the rule {Milk → Bread}.				4
2	An online bookstore discovers that customers who buy "Data Science" books often buy "Python Programming" books. Scenario Question: Explain how lift can help determine whether this association is meaningful				4
3	Amazon suggests products like "Phone Cases" after a customer buys a smartphone. Scenario Question: Identify whether this is collaborative filtering or content-based filtering and justify your answer				4
4	A new e-commerce site wants to implement a recommendation engine but has limited user data. Scenario Question: Suggest which filtering technique is more suitable initially and why.				4
5	A streaming service applies PCA (dimensionality reduction) before collaborative filtering. Scenario Question: Explain why dimensionality reduction might improve recommendation performance.				4
6	A video platform uses user ratings to recommend similar movies. Scenario Question: Describe how user-item interaction matrices are used in collaborative filtering.				4
7	A grocery store wants to recommend items to customers based on frequent co-purchases. Scenario Question: Discuss how association rules can be integrated into a recommendation engine.				4
8	Netflix recommends movies to users based on their viewing history. Scenario Question: Explain how collaborative filtering works in this case.				
9	A retail chain uses Apriori algorithm to generate rules. One rule is {Diapers → Beer}. Scenario Question: Discuss why this rule might be considered strong or weak depending on support and confidence values.				4
10	A supermarket analyzes customer transactions and finds that many customers who buy bread and butter also buy jam. Question: Identify the association rule from the above scenario. Explain how association rule mining helps supermarkets improve sales strategies.				4
11	A retail store records the following purchases: • Total transactions = 200 • Transactions containing Laptop = 80 • Transactions containing Mouse = 60 • Transactions containing Laptop and Mouse = 50 Question: Calculate the Support, Confidence, and explain what these values indicate for the rule Laptop → Mouse.				4
12	Smart Traffic System Scenario A smart city uses an intelligent traffic signal system that observes vehicle density and decides when to change signals to reduce congestion. Question: Explain how reinforcement learning can be used in this system. Identify the agent, environment, actions, states, and rewards.				6
13	A warehouse robot is trained using Reinforcement Learning to deliver packages. It has already discovered one route that takes **8 minutes** to reach the destination. However, there may be **other routes that could be faster**, but the robot has never tried them. **Questions:** 1. Explain how **exploration** and **exploitation** apply in this situation. 2. What might happen if the robot only uses **exploitation**? 3. Why is balancing exploration and exploitation important for the robot?				6

14	A robot vacuum cleaner learns to avoid obstacles by trial and error. Question: Explain how this scenario demonstrates the definition of Reinforcement Learning, highlighting agent, environment, actions, and rewards.	6
15	A Reinforcement Learning system is used to recommend treatments for patients. The system has identified a **treatment that works well for most patients**, but new treatments are constantly being developed. **Questions:** 1. Identify the **exploration** and **exploitation** actions in this scenario. 2. What are the risks if the system performs **too much exploration**? 3. Explain why a balance between exploration and exploitation is important in healthcare.	6
16	A healthcare system uses reinforcement learning to recommend personalized treatments. Question: Analyze how exploration vs exploitation impacts patient safety and treatment effectiveness.	6
17	A bank uses Reinforcement Learning to detect fraudulent transactions. The system has learned certain **patterns that indicate fraud** and usually blocks transactions based on these patterns. However, fraudsters constantly develop **new strategies**. **Questions:** 1. Explain how **exploration** helps the system detect new fraud patterns. 2. Explain how **exploitation** helps in quickly blocking known fraud activities. 3. Why must the bank maintain a balance between exploration and exploitation?	6
18	A food delivery app must decide whether to recommend a popular restaurant (exploitation) or a new one (exploration). Question: Analyze the exploration vs exploitation dilemma in this case and suggest a balanced strategy.	6